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How informative is the initial condition in the dynamic panel model with fixed effects?

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Abstract

I consider estimation of the autoregressive panel model with fixed effects $y_{it} = c_i + \beta y_{i,t-1} + \varepsilon_{it}$. I investigate the estimation method developed by Blundell and Bond (1998), which makes use of the stationarity of the initial levels. I do it by numerically comparing the semiparametric information bounds for the case that incorporates the stationarity of the initial condition and for the case that does not. It is found that the efficiency gain can be substantial. © 1999 Elsevier Science S.A. All rights reserved.

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1. Introduction

The purpose of this paper is to examine the role of initial condition in a dynamic panel model with fixed effects as a potential source of efficiency gain. I consider efficient semiparametric estimation of the autoregressive parameter

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β in a dynamic panel model with fixed effects

$$y_{it} = c_i + \beta y_{i,t-1} + \varepsilon_{it}, \quad t = 1, 2, \ldots, T; \, i = 1, \ldots, n,$$

where n is relatively large, and T is moderately small. I make a standard sequential conditional moment restriction that

$$\mathrm{E}[\varepsilon_t \,|\, c, y_0, \ldots, y_{t-1}] = 0, \quad t = 1, 2, \ldots, T \tag{1}$$

(I omit the i subscript whenever obvious.) Nonparametric component of this model is the distribution of ε_ts and c.

It is fairly common in the panel literature to disregard the potentially informative role of the distribution of y_0 for the estimation of β. This practice is understandable because misspecification of the distribution of y_0 would result in the inconsistency of the resultant estimator. Perhaps because of this concern, efficiency in dynamic panel literature has been discussed in the framework where y_0 was assumed to be ancillary for the parameter of interest. Various efficient estimators have been developed in this limited context. Holtz-Eakin et al. (1988), Arellano and Bond (1991), and Hahn (1997) developed linear GMM estimators, which use the lagged levels of the series as instruments to the first differences. Ahn and Schmidt (1995) discovered additional (nonlinear) moment conditions for efficiency gain.

Recently, Blundell and Bond (1998) suggested an alternative linear GMM estimator based on additional moment restrictions, which are valid if we have

$$y_0 = \frac{c}{1-\beta} + u = \frac{c}{1-\beta} + \sum_{t=0}^{\infty} \beta^t \varepsilon_{-t}, \tag{2}$$

where ε_t are i.i.d. mean-zero random variables. Observe that the stationarity assumption (2) is not implied by the sequential conditional moment restrictions (1). In their Monte Carlo study, they report good finite sample properties. The purpose of this paper is to investigate the source of the efficiency gain by comparing the semiparametric information bound for the case which makes use of the initial condition with the one which does not: I do *not* compare a particular estimator with another.

To compute the additional information due to the stationarity (2), it would be ideal to have a semiparametric information bound in the model (1). Unfortunately, such a bound does not exist in the literature. The only available bound in the literature is Chamberlain's (1992) bound, which is based on

$$\mathrm{E}[\varepsilon_t \,|\, y_0, \ldots, y_{t-1}] = 0, \quad t = 1, 2, \ldots, T. \tag{3}$$

It is obvious that (3) does not imply (1): Although the moment restrictions

$$\mathrm{E}[y_{t-2}(\varepsilon_t - \varepsilon_{t-1})] = \cdots = \mathrm{E}[y_0(\varepsilon_t - \varepsilon_{t-1})] = 0, \quad t = 2, 3, \ldots, T \tag{4}$$

used by the usual linear GMM estimator are implied by (3), the nonlinear moment restriction

$$\mathrm{E}[(c + \varepsilon_T)(\varepsilon_t - \varepsilon_{t-1})] = 0, \quad t = 2, \ldots, T-1 \tag{5}$$

used by Ahn and Schmidt (1995) does not follow as a consequence of (3).

Because the semiparametric information bound for (1) is not available in the literature, and because it is most convenient to impose stationarity on initial levels when the error ε_t is i.i.d. normal, I assume that the DGP follows

$$\varepsilon_t | c, y_0, \ldots, y_{t-1} \sim \mathrm{N}(0, \sigma^2), \quad t = 1, 2, \ldots, T \tag{6}$$

and

$$y_0 | c \sim \mathrm{N}\left(\frac{c}{1-\beta}, \frac{\sigma^2}{1-\beta^2}\right). \tag{7}$$

The model is still semiparametric because the distribution of c is not parametrically specified. I compute the semiparametric information bound for β under (6) only, and the one under (6) and(7). Adding information clearly cannot hurt, and therefore, the latter information is expected to be strictly larger than the former on theoretical grounds.

In order to quantify the marginal information due to (7), I assume that c has a normal distribution, and numerically evaluate the bounds under various parameter combinations. One finds that the efficiency gain can be quite substantial even when the observation per individual $T+1$ is moderately big.[1] To compare the stationarity related efficiency gain with the nonlinear-moment-restriction related efficiency gain, I numerically evaluate Chamberlain's (1992) semiparametric information bound under the same parameter combinations. Numerical comparison reveals that the stationarity related efficiency gain substantially dominates the other.

[1] This exercise is similar in spirit to Ahn and Schmidt (1995), who evaluated the efficiency gain due to the nonlinear-moment-restriction by numerically comparing the asymptotic variances of various estimators.

The layout of the paper is as follows. In Section 2, I briefly review the standard moment restrictions for the autoregressive panel model with fixed effects. In Section 3, I compute the semiparametric information bounds under (6) only, and under (6) along with (7). In Section 4, I present various information bounds under various parameter combinations. Section 5 concludes.

2. Semiparametric information bounds

In order to assess the marginal contribution of stationarity assumption (2), it is useful to compare the marginal information in the stationarity assumption in relation to the standard sequential moment restrictions. Information in a panel model with fixed effects is by definition semiparametric information: Distribution of the fixed effects is nonparametrically specified.

I focus on the case where semiparametric information bounds can be computed. I thus make an important restrictive assumption regarding the data generating process. I assume that

Assumption 1. $\varepsilon_t | y_{t-1}, \ldots, y_0, c \sim \mathrm{N}(0, \sigma^2)$, $t = 1, \ldots, T$, where σ^2 is unknown.

Assumption 2. $y_0 | c \sim \mathrm{N}(\frac{c}{1-\beta}, \frac{\sigma^2}{1-\beta^2})$.

Observe that the model is semiparametric even with Assumption 2 because the distribution of the individual specific effect c is nonparametrically specified.

To assess the marginal contribution of the stationarity assumption, I first compute the semiparametric information bound assuming that the stationarity is not part of our knowledge even though the data generating process follows the stationarity assumption. Under this assumption, we have another nonparametric component in the information calculation. We now have the distribution of y_0 nonparametrically specified. Furthermore, y_0 is ancillary β because the distribution of y_0 does not exhibit any dependence on β under our ignorance of the stationarity. For an excellent survey on semiparametric information bound, see Bickel et al. (1993) or Newey (1990). Semiparametric information bound is the expected value of the square of the efficient score.

Theorem 1. Under Assumption 1, the efficient score for β equals

$$\ell_\beta^* - \frac{\mathrm{E}[\ell_\beta^* \ell_{\sigma^2}^*]}{\mathrm{E}[(\ell_{\sigma^2}^*)^2]}\, \ell_{\sigma^2}^*,$$

where

$$\ell^*_\beta = -\frac{1}{\sigma^2}\left(\sum_{t=1}^{T-1} y_t - \mathrm{E}\left[\sum_{t=1}^{T-1} y_t \middle| y_0, S^*\right]\right)\mathrm{E}[c\,|\,y_0, S^*]$$

$$+\frac{1}{\sigma^2}\left(\sum_{t=1}^{T} y_{t-1}(y_t-\beta y_{t-1}) - \mathrm{E}\left[\sum_{t=1}^{T} y_{t-1}(y_t-\beta y_{t-1})\middle| y_0, S^*\right]\right), \quad (8)$$

$$\ell^*_{\sigma^2} = \frac{1}{2(\sigma^2)^2}\left(\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2 - \mathrm{E}\left[\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2\middle| y_0, S^*\right]\right) \quad (9)$$

and

$$S^* \equiv \sum_{t=1}^{T}(y_t-\beta y_{t-1}).$$

Proof. See Appendix A. □

Now assume that the stationarity is part of our knowledge. We then know that Assumption 2 holds. Notice that y_0 is no longer ancillary. Furthermore, the only nonparametric component under this situation is the distribution of c.

Theorem 2. Under Assumptions 1 and 2, the efficient score for β equals

$$\ddot{\ell}_\beta - \frac{\mathrm{E}[\ddot{\ell}_\beta\ddot{\ell}_{\sigma^2}]}{\mathrm{E}[(\ddot{\ell}_{\sigma^2})^2]}\,\ddot{\ell}_{\sigma^2},$$

where

$$\ddot{\ell}_\beta = -\frac{1}{\sigma^2}\left(\sum_{t=1}^{T-1} y_t - \mathrm{E}\left[\sum_{t=1}^{T-1} y_t \middle| \ddot{S}\right]\right)\mathrm{E}[c\,|\,\ddot{S}]$$

$$+\frac{1}{\sigma^2}\left(\sum_{t=1}^{T}(y_t-\beta y_{t-1})y_{t-1} - \mathrm{E}\left[\sum_{t=1}^{T}(y_t-\beta y_{t-1})y_{t-1}\middle| \ddot{S}\right]\right)$$

$$+\frac{\beta}{\sigma^2}(y_0^2 - \mathrm{E}[y_0^2|\ddot{S}]), \quad (10)$$

$$\ddot{\ell}_{\sigma^2} = \frac{1}{2(\sigma^2)^2}\left(\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2 - \mathrm{E}\left[\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2\middle|\ddot{S}\right]\right)$$
$$+\frac{1-\beta^2}{2(\sigma^2)^2}(y_0^2-\mathrm{E}[y_0^2\,|\,\ddot{S}]) \tag{11}$$

and

$$\ddot{S} \equiv \sum_{t=1}^{T}(y_t-\beta y_{t-1}) + (1+\beta)y_0.$$

Proof. See Appendix B. □

It is easy to conjecture that the information bound in Theorem 2 is bigger than the one in Theorem 1. For a formal proof that it is indeed so, see Appendix C.

3. Numerical comparison

To assess the efficiency gain quantitatively, we need to condition on a particular distribution on c. To simplify the analytic calculation, I further assume that

Assumption. $c \sim \mathrm{N}(0, \omega^2)$.

The fact that the fixed effects follow a normal distribution is assumed unknown to an econometrician. Although this data generating process is restrictive due to the homoscedasticity and normality assumption, the quantitative result is at least suggestive that the initial condition can be quite informative about β. I then go on to compute the ratio of information bounds computed in Theorems 1 and 2. Such comparison would be meaningful only if both bounds are achievable.[2] It turns out that both are. Pfanzagl (1990) considered general mixture models, and suggested a sample-splitting-based efficient estimator construction strategy.

Given that the calculations are performed for normal c, comparisons with the Cramer–Rao lower bound for the parametric model with normal c may be deemed useful. Such comparisons do not seem to be necessary. Suppose first that stationarity is not assumed. The Fisher information in the parametric model with normal c should be greater than or equal to the one in the

[2] I am grateful to Oliver Linton for pointing this out.

Table 1
Stationarity-related efficiency gain ($T = 3$)

β	ω^2/σ^2 0	0.25	0.5	1	2	4
− 0.99	1.12	5.20	12.59	30.18	69.42	156.46
− 0.9	1.08	1.54	2.40	4.72	11.11	30.17
− 0.8	1.05	1.30	1.77	3.16	7.41	21.42
− 0.5	1.01	1.09	1.27	1.88	3.97	11.56
− 0.3	1.03	1.05	1.13	1.43	2.51	6.50
0	1.13	1.10	1.09	1.08	1.08	1.08
0.3	1.48	1.79	2.40	4.52	12.34	42.15
0.5	2.40	5.21	10.75	30.26	102.65	380.46
0.8	20.09	218.66	649.14	2207.43	8109.04	31 047.41
0.9	106.44	3472.83	11 371.84	40 550.06	152 253.32	588 886.27
0.99	14 462.97	0.3×10^8	0.1×10^9	0.4×10^9	0.1×10^{10}	0.6×10^9

semiparametric model derived in Theorem 1. On the other hand, the information in the semiparametric models should be greater than or equal to the one in models where the only restriction is given in the expression of (unconditional) second moments of $y_0, \ldots, y_T$ as functions of underlying parameters. Now, the information in models with second moment restriction should be equal to the information in the parametric models with normally distributed fixed effects because second moments exhausts information in general normal models. Similar reasoning applies to the case where stationarity is assumed. Therefore, the semiparametric information bounds in Theorems 1 and 2 will in fact coincide with the Cramer–Rao lower bounds for the particular DGP I consider.[3]

The numbers in Tables 1–3 are the ratio of information bounds computed in Theorems 1 and 2. I picked the parameter combinations according to the choice in Ahn and Schmidt (1995). It is clear that the marginal information contained in the initial condition is substantial even when the number of observations per individual is relatively large. We find some interesting patterns. First, the marginal gain tends to be larger for $|\beta|$ close to one. Intuitively, this can be explained by the alternative expression of y_t

$$y_t = \beta^t y_0 + \frac{1-\beta^t}{1-\beta}\, c + \sum_{j=0}^{t-1} \beta^j \varepsilon_{t-j}.$$

[3] In this sense, parametric models with normally distributed fixed effects are expected to be the 'worst' parametric models: Models with normally distributed fixed effects contain the least information. I am grateful to Gary Chamberlain for providing me with this insight.

Table 2
Stationarity-related efficiency gain ($T = 4$)

β	ω^2/σ^2 0	0.25	0.5	1	2	4
− 0.99	1.00	4.15	9.47	21.65	48.25	106.72
− 0.9	1.00	1.34	1.96	3.59	8.02	21.12
− 0.8	1.00	1.17	1.52	2.52	5.50	15.29
− 0.5	1.01	1.06	1.19	1.62	3.08	8.34
− 0.3	1.01	1.03	1.09	1.29	2.01	4.66
0	1.07	1.05	1.04	1.04	1.03	1.03
0.3	1.28	1.43	1.75	2.87	6.94	22.35
0.5	1.80	3.18	5.87	15.14	49.09	178.37
0.8	10.75	99.06	279.13	911.37	3260.49	12 294.00
0.9	54.89	1508.28	4657.70	15 879.71	57 926.95	220 363.77
0.99	7664.72	0.1×10^8	0.4×10^8	0.2×10^9	0.6×10^9	0.2×10^{10}

The coefficient β^t of y_0 indicates that the importance of initial condition in y_t is an increasing function of $|\beta|$.[4] Second, the marginal gain seen as a function of β is asymmetric around zero. Comparing the $\beta = 0.8$ case with the $\beta = -0.8$ case reveals that the marginal gain is larger when β is positive rather than negative. This is probably because of the differential effects of β on the conditional expectation and variance of y_0 given c. Observe that

$$y_0|c \sim \mathrm{N}\left(\frac{c}{1-\beta}, \frac{\sigma^2}{1-\beta^2}\right).$$

The conditional expectation $c/(1-\beta)$ increases in absolute value as β increases, whereas the conditional variance $\sigma^2/(1-\beta^2)$ increases as $|\beta|$ increases. It therefore seems natural that the marginal effect of β is the bigger when it is positive. Third, the marginal gain tends to be small when ω is small relative to σ. This could probably be explained by the same intuition as before. When ω is small, the fixed effect c is close to being constant, and the conditional expectation $c/(1-\beta)$ of y_0 given c would have a small variation. Hence, the effect of stationarity via the conditional expectation would be small. On the other hand, the conditional variance $\sigma^2/(1-\beta^2)$ does not depend on c, and the effect of stationarity via the conditional variance would still remain. Therefore, even when $\omega = 0$, the stationarity implies some marginal gain.

[4] I am grateful to an anonymous referee for providing me with this intuition.

Table 3
Stationarity-related efficiency gain ($T = 10$)

	ω^2/σ^2					
β	0	0.25	0.5	1	2	4
− 0.99	1.00	2.79	5.22	10.35	21.15	44.60
− 0.9	1.00	1.19	1.46	2.13	3.89	9.04
− 0.8	1.00	1.10	1.25	1.64	2.80	6.57
− 0.5	1.00	1.03	1.08	1.24	1.77	3.67
− 0.3	1.00	1.01	1.03	1.11	1.35	2.26
0	1.01	1.00	1.00	1.00	1.00	1.00
0.3	1.05	1.08	1.16	1.43	2.38	5.98
0.5	1.14	1.40	1.90	3.58	9.67	32.66
0.8	2.41	11.91	29.84	90.69	312.17	1154.14
0.9	8.05	132.14	369.57	1180.05	4142.32	15 431.30
0.99	1120.29	907 028.06	0.3×10^7	0.9×10^7	0.3×10^8	0.1×10^9

Table 4
Nonlinear-moment-restriction-related efficiency gain ($T = 3$)

	ω^2/σ^2					
β	0	0.25	0.5	1	2	4
− 0.99	1.00	1.01	1.01	1.01	1.01	1.01
− 0.9	1.05	1.05	1.06	1.07	1.09	1.12
− 0.8	1.08	1.09	1.11	1.14	1.18	1.25
− 0.5	1.14	1.20	1.26	1.35	1.51	1.71
− 0.3	1.15	1.25	1.35	1.51	1.77	2.11
0	1.14	1.31	1.49	1.80	2.27	2.86
0.3	1.11	1.39	1.67	2.18	2.95	3.91
0.5	1.08	1.45	1.84	2.53	3.58	4.87
0.8	1.03	1.64	2.26	3.34	4.96	6.97
0.9	1.01	1.76	2.49	3.75	5.62	7.96
0.99	1.00	1.94	2.78	4.21	6.35	9.03

To assess the efficiency gain due to the initial condition in perspective, I also compute the ratios of information bound in Theorem 1 with the one computed by Chamberlain's (1992) and summarized them in Tables 4–6. It is important to note that Chamberlain's (1992) bound is based only on the moment restriction (3): Normality is not part of his model. Observe that, although the efficiency gain is substantial, it is in general tiny compared with the information contained in the initial condition. On the other hand, the gain due to stationarity is rather small for small values of $|\beta|$.

Table 5
Nonlinear-moment-restriction-related efficiency gain ($T = 4$)

β	ω^2/σ^2 0	0.25	0.5	1	2	4
− 0.99	1.00	1.00	1.00	1.00	1.01	1.01
− 0.9	1.03	1.04	1.04	1.05	1.07	1.10
− 0.8	1.06	1.07	1.09	1.11	1.15	1.21
− 0.5	1.12	1.12	1.23	1.30	1.41	1.55
− 0.3	1.15	1.25	1.34	1.47	1.64	1.84
0	1.17	1.37	1.53	1.78	2.10	2.43
0.3	1.16	1.50	1.79	2.23	2.79	3.35
0.5	1.13	1.61	2.03	2.66	3.46	4.26
0.8	1.06	1.89	2.62	3.74	5.12	6.50
0.9	1.03	2.07	2.96	4.31	6.00	7.67
0.99	1.00	2.33	3.41	5.02	7.03	9.04

Table 6
Nonlinear-moment-restriction-related efficiency gain ($T = 10$)

β	ω^2/σ^2 0	0.25	0.5	1	2	4
− 0.99	1.00	1.00	1.00	1.00	1.00	1.01
− 0.9	1.02	1.03	1.03	1.03	1.04	1.05
− 0.8	1.04	1.05	1.06	1.07	1.08	1.10
− 0.5	1.07	1.12	1.15	1.78	1.21	1.25
− 0.3	1.10	1.18	1.22	1.27	1.32	1.37
0	1.15	1.30	1.38	1.47	1.55	1.62
0.3	1.19	1.49	1.65	1.81	1.94	2.05
0.5	1.21	1.71	1.96	2.22	2.43	2.57
0.8	1.17	2.42	3.06	3.71	4.22	4.56
0.9	1.11	2.96	3.92	4.89	5.65	6.16
0.99	1.01	3.94	5.42	6.89	8.07	8.86

Although the marginal gain due to stationarity can be substantial, it cannot be too much emphasized that such gain is feasible only under correct specification. Violation of stationarity assumption would lead to inconsistency for most estimators, and the resultant bias-variance trade-offs in finite samples would necessitate some decision scientific choice. Analysis of such trade-offs in finite samples is possible only for a given set of estimators, which is beyond the scope of this paper. In any case, the numbers in Tables 1–6 indicate that *correct* specification of the distribution of initial condition can in principle lead to a substantial efficiency gain.

We have an interesting by-product in this numerical exercise. We can assess the marginal information due to conditioning on c as in (1). Also, we can conclude that Ahn and Schmidt's (1995) additional moment restriction (5) virtually exhausts all the marginal information due to such conditioning. This can be explained as follows. Under the particular DGP that I consider, it can analytically be shown that the linear IV approach as used by Arellano and Bond (1991) achieves Chamberlain's (1992) bound. Therefore, under my particular DGP, Ahn and Schmidt's (1995) Table 1, which compares the information contained in Arellano and Bond's (1991) estimator with the one contained in theirs, can be reinterpreted as a comparison of the information in the moment restriction (3) with the one contained in both (3) and (5). Comparison of Ahn and Schmidt's (1995) Table 1 with Tables 4–6 reveals that the stronger form of the sequential moment restriction (1) is not too informative compared with the two moment restrictions (3) and (5). We can therefore conclude that Ahn and Schmidt's (1995) estimator is almost efficient in the sense that it almost achieves the semiparametric information bound for the model (1).

4. Summary

In this paper, I computed the information bounds for panel autoregressive model with fixed effects with and without incorporating the distribution of initial condition implied by stationarity assumption. Numerical comparison under various parameter combination suggests that the efficiency gain due to the stationarity assumption can be quite substantial. This result suggests that future research may be more productive if more attention is given to the stationarity assumption or initial condition in general.

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Appendix A. Proof of Theorem 1

Drop the i subscript, and calculate the conditional density given (y_0, c)

$$p(y_T, \ldots, y_1 \mid c, y_0; \beta, \sigma^2) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^T \exp\left[-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(y_t - c - \beta y_{t-1})^2\right].$$

Observe that

$$\log p(y_T, \dots, y_1 \mid c, y_0; \beta, \sigma^2)$$

$$= \frac{T}{2} \log \sigma^2 + \frac{T}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{t=1}^{T} (y_t - c - \beta y_{t-1})^2$$

$$= \frac{\sum_{t=1}^{T} (y_t - \beta y_{t-1})}{\sigma^2} c - \frac{T}{2\sigma^2} c^2 + \frac{T}{2} \log \sigma^2 + \frac{T}{2} \log(2\pi)$$

$$- \frac{1}{2\sigma^2} \sum_{t=1}^{T} (y_t - \beta y_{t-1})^2.$$

Tentatively assume that (β, σ^2) is known, and treat c as a 'parameter'. We then recognize that $p(y_T, \dots, y_1 \mid c, y_0; \beta, \sigma^2)$ form an exponential family with a complete sufficient statistic $(\sum_{t=1}^{T} (y_t - \beta y_{t-1}))/\sigma^2$. Our information bound calculation is based on this observation.

Now, compute the conditional density given y_0

$$p(y_T, \dots, y_1 \mid y_0; \beta, \sigma^2)$$

$$= \int p(y_T, \dots, y_1 \mid c, y_0; \beta, \sigma^2) \, \mathrm{d}\Gamma(c \mid y_0)$$

$$= \int \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^T \exp\left[- \frac{1}{2\sigma^2} \sum_{t=1}^{T} (y_t - c - \beta y_{t-1})^2 \right] \mathrm{d}\Gamma(c \mid y_0), \tag{A.1}$$

where $\Gamma(\cdot \mid y_0)$ denotes the conditional distribution of c given y_0.

Observe that the log-likelihood given (y_0, c) of a parametric submodel equals[5]

$$\log p(y_T, \dots, y_1 \mid c, y_0; \beta, \sigma^2, \delta)$$

$$= \frac{c}{\sigma^2} \sum_{t=1}^{T} (y_t - \beta y_{t-1}) - \frac{1}{2\sigma^2} \sum_{t=1}^{T} (y_t - \beta y_{t-1})^2 - \frac{Tc^2}{2\sigma^2} - \frac{T}{2} \log \sigma^2$$

$$+ \log g(c \mid y_0, \delta). \tag{A.2}$$

[5] Ignore the constant $-(T/2)\log(2\pi)$.

Here, $g(c \mid y_0, 0) = \gamma(c \mid y_0)$, where $\gamma(c \mid y_0)$ is the density of $\Gamma(c \mid y_0)$. Note that (A.2) corresponds to the parametric submodel of (A.1):

$$p_G(y_T, \dots, y_1 \mid y_0; \beta, \sigma^2, \delta)$$

$$= \int \left(\frac{1}{\sigma\sqrt{2\pi}}\right)^T \exp\left[-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(y_t - c - \beta y_{t-1})^2\right] g(c \mid y_0, \delta)\,\mathrm{d}c. \tag{A.3}$$

Calculation of Scores. Let $\tilde{\ell}_\theta = \partial \log p_G / \partial\theta$. We then have

$$\tilde{\ell}_\beta = -\frac{c}{\sigma^2}\sum_{t=1}^{T} y_{t-1} + \frac{1}{\sigma^2}\sum_{t=1}^{T} y_{t-1}(y_t - \beta y_{t-1}),$$

$$\tilde{\ell}_{\sigma^2} = -\frac{c}{(\sigma^2)^2}\sum_{t=1}^{T}(y_t - \beta y_{t-1}) + \frac{1}{2(\sigma^2)^2}\sum_{t=1}^{T}(y_t - \beta y_{t-1})^2 + \frac{Tc^2}{2(\sigma^2)^2} - \frac{T}{2\sigma^2},$$

$$\tilde{\ell}_\delta = \frac{g_\delta(c \mid y_0, 0)}{g(c \mid y_0, 0)}. \tag{A.4}$$

Getting back to (A.3), we have the scores by Ibragimov and Has'minskii (1981)

$$\ell = \mathrm{E}[\tilde{\ell} \mid y_0, y_1, \dots, y_T],$$

where

$$\ell = (\ell_\beta\ \ell_{\sigma^2}\ \ell_\delta)' \quad \text{and} \quad \tilde{\ell} = (\tilde{\ell}_\beta\ \tilde{\ell}_{\sigma^2}\ \tilde{\ell}_\delta)'.$$

Because of the sufficient statistic structure, for any measurable function $f(c)$, we have

$$\mathrm{E}[f(c) \mid y_0, y_1, \dots, y_T] = \mathrm{E}[f(c) \mid y_0, S^*], \tag{A.5}$$

where

$$S^* = \sum_{t=1}^{T}(y_t - \beta y_{t-1}).$$

Thus, we have

$$\ell_\beta = -\frac{\sum_{t=1}^T y_{t-1}}{\sigma^2}\,\mathrm{E}[c\,|\,y_0,\,S^*] + \frac{1}{\sigma^2}\sum_{t=1}^T y_{t-1}(y_t - \beta y_{t-1}),$$

$$\ell_{\sigma^2} = -\frac{S^*}{(\sigma^2)^2}\,\mathrm{E}[c\,|\,y_0,\,S^*] + \frac{1}{2(\sigma^2)^2}\sum_{t=1}^T (y_t - \beta y_{t-1})^2$$

$$+ \frac{T}{2(\sigma^2)^2}\,\mathrm{E}[c^2\,|\,y_0,\,S^*] - \frac{T}{2\sigma^2}.$$

Now, the nonparametric tangent space $\mathbf{K}$ is calculated. From (A.4) and (A.5), it follows that $\mathbf{K}$ is the closure of the space consisting of functions of the form

$$(y_0,\,y_1,\dots,\,y_T) \mapsto \frac{\int k(c,\,y_0)p_0(S^*(y_0,\,y_1,\dots,\,y_T,\,\beta),\,c)\,\mathrm{d}\Gamma(c\,|\,y_0)}{\int p_0(S^*(y_0,\,y_1,\dots,\,y_T,\,\beta),\,c)\,\mathrm{d}\Gamma(c\,|\,y_0)},$$

where

$$\int k(c,\,y_0)\,\mathrm{d}\Gamma(c\,|\,y_0) = 0.$$

Here, $p_0(S^*(y_0,\,y_1,\dots,\,y_T,\,\beta),\,c)$ denotes the conditional density of $S^*(y_0,\,y_1,\dots,\,y_T,\,\beta)$ for some fixed y_0 given c. Let $\mathbf{K}_{y_0}$ denote the closure of the space consisting of functions of the form

$$(y_1,\dots,\,y_T) \mapsto \frac{\int k(c,\,y_0)p_0(S^*(y_0,\,y_1,\dots,\,y_T,\,\beta),\,c)\,\mathrm{d}\Gamma(c\,|\,y_0)}{\int p_0(S^*(y_0,\,y_1,\dots,\,y_T,\,\beta),\,c)\mathrm{d}\Gamma(c\,|\,y_0)},$$

where

$$\int k(c,\,y_0)\,\mathrm{d}\Gamma(c\,|\,y_0) = 0.$$

It will be shown that

$$\left\{s \mapsto \frac{\int k(c,\,y_0)p_0(s,\,c)\,\mathrm{d}\Gamma(c\,|\,y_0)}{\int p_0(s,\,c)\,\mathrm{d}\Gamma(c\,|\,y_0)} :\ \int k(c,\,y_0)\,\mathrm{d}\Gamma(c\,|\,y_0) = 0\right\} = L_*(R_{\Gamma,y_0}), \quad \text{(A.6)}$$

where $L_*(R_{\Gamma,y_0})$ denotes the set of zero mean functions in $L_2(R_{\Gamma,y_0})$. Here, R_{Γ,y_0} denotes the conditional distribution of S^* given y_0. Because $\subset$ is obvious, only the other implication is proved. Assume that $h(s) \in L_*(R_{\Gamma,y_0})$ is orthogonal to

$$s \mapsto \frac{\int k(c,\, y_0) p_0(s,\, c)\, \mathrm{d}\Gamma(c \mid y_0)}{\int p_0(s,\, c)\, \mathrm{d}\Gamma(c \mid y_0)}.$$

Then,

$$\begin{aligned}
&\int h(s) \cdot \frac{\int k(c,\, y_0) p_0(s,\, c)\, \mathrm{d}\Gamma(c \mid y_0)}{\int p_0(s,\, c)\, \mathrm{d}\Gamma(c \mid y_0)}\, R_{\Gamma,y_0}(\mathrm{d}s) \\
&\quad = \int k(c,\, y_0) \left[\int h(s) p_0(s,\, c)\, \mathrm{d}s \right] \mathrm{d}\Gamma(c \mid y_0) \\
&\quad = 0.
\end{aligned}$$

If this holds for all $k(c,\, y_0)$, then

$$\Gamma \left\{ c\colon \int h(s) p_0(s,\, c)\, \mathrm{d}s = 0 \right\} = 1.$$

But because S^* is complete and sufficient statistic for c, we should have $h = 0$. (A.6) is thus proved.

From (A.6), it follows that the projection on $\mathbf{K}$ is equivalent to the conditional expectation given S^* and y_0. The efficient score is the residual in the projection.

Appendix B. Proof of Theorem 2

From the likelihood of the parametric submodel of $y_0, y_1, \ldots, y_T, c$ under consideration

$$\left(\frac{1}{\sigma\sqrt{2\pi}} \right)^T \exp\left[-\frac{1}{2\sigma^2} \sum_{t=1}^{T} (y_t - c - \beta y_{t-1})^2 \right] \times \frac{\sqrt{1-\beta^2}}{\sigma\sqrt{2\pi}}$$

$$\exp\left[-\frac{1-\beta^2}{2\sigma^2} \left(y_0 - \frac{c}{1-\beta} \right)^2 \right],$$

which delivers the log-likelihood (again, ignoring irrelevant constants)

$$\frac{1}{2}\log(1-\beta^2)-\frac{T+1}{2}\log\sigma^2+\frac{\sum_{t=1}^{T}(y_t-\beta y_{t-1})+(1+\beta)y_0}{\sigma^2}c$$

$$-\frac{1}{2\sigma^2}\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2-\frac{1-\beta^2}{2\sigma^2}y_0^2-\frac{1}{2\sigma^2}\left(T+\frac{1+\beta}{1-\beta}\right)c^2+\log g(c,\delta)$$

,

we obtain

$$\tilde{\ell}_\beta=-\frac{\beta}{1-\beta^2}+\frac{-\sum_{t=1}^{T}y_{t-1}+y_0}{\sigma^2}c$$

$$+\frac{1}{\sigma^2}\sum_{t=1}^{T}(y_t-\beta y_{t-1})y_{t-1}+\frac{\beta}{\sigma^2}y_0^2-\frac{c^2}{\sigma^2(1-\beta)^2},$$

$$\tilde{\ell}_{\sigma^2}=-\frac{T+1}{2\sigma^2}-\frac{\sum_{t=1}^{T}(y_t-\beta y_{t-1})+(1+\beta)y_0}{(\sigma^2)^2}c+\frac{1}{2(\sigma^2)^2}\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2$$

$$+\frac{1-\beta^2}{2(\sigma^2)^2}y_0^2+\frac{1}{2(\sigma^2)^2}\left(T+\frac{1+\beta}{1-\beta}\right)c^2$$

and

$$\ell_\beta=\mathrm{E}[\tilde{\ell}_\beta\,|\,y_0,y_1,\dots,y_T]$$

$$=-\frac{\beta}{1-\beta^2}+\frac{-\sum_{t=1}^{T}y_{t-1}+y_0}{\sigma^2}\mathrm{E}[c|\ddot{S}]$$

$$+\frac{1}{\sigma^2}\sum_{t=1}^{T}(y_t-\beta y_{t-1})y_{t-1}+\frac{\beta}{\sigma^2}y_0^2-\frac{1}{\sigma^2(1-\beta)^2}\mathrm{E}[c^2\,|\,\ddot{S}],$$

$$\ell_{\sigma^2}=\mathrm{E}[\tilde{\ell}_{\sigma^2}\,|\,y_0,y_1,\dots,y_T]$$

$$=-\frac{T+1}{2\sigma^2}-\frac{\ddot{S}}{(\sigma^2)^2}\mathrm{E}[c\,|\,\ddot{S}]+\frac{1}{2(\sigma^2)^2}\sum_{t=1}^{T}(y_t-\beta y_{t-1})^2+\frac{1-\beta^2}{2(\sigma^2)^2}y_0^2$$

$$+\frac{1}{2(\sigma^2)^2}\left(T+\frac{1+\beta}{1-\beta}\right)\mathrm{E}[c^2\,|\,\ddot{S}]$$

for

$$\ddot{S} = \sum_{t=1}^{T} (y_t - \beta y_{t-1}) + (1+\beta) y_0.$$

By the same reasoning as in the previous section, we obtain the efficient score by

$$\ddot{\ell}_\beta = \ell_\beta - \mathrm{E}[\ell_\beta \mid \ddot{S}], \qquad \ddot{\ell}_{\sigma^2} = \ell_{\sigma^2} - \mathrm{E}[\ell_{\sigma^2} \mid \ddot{S}].$$

Appendix C. Theoretical comparison of bounds

Observe that

$$\mathrm{E}[f(c) \mid \ddot{S}] = E[(c) \mid y_0, S^*]$$

for any measurable function $f(\cdot)$ if

$$y_0 | c \sim \mathrm{N}\left(\frac{c}{1-\beta}, \frac{\sigma^2}{1-\beta^2}\right). \tag{C.1}$$

This is due to the sufficiency of $\ddot{S}$ for c under (C.1). With this in mind, we can rewrite

$$\begin{bmatrix} \ell_\beta^* \\ \ell_{\sigma^2}^* \end{bmatrix} = \begin{bmatrix} (\varphi_1 - \mathrm{E}[\varphi_1 \mid y_0, S^*])\mathrm{E}[c \mid \ddot{S}] + (\varphi_2 - \mathrm{E}[\varphi_2 \mid y_0, S^*]) \\ \varphi_3 - \mathrm{E}[\varphi_3 \mid y_0, S^*] \end{bmatrix},$$

$$\begin{bmatrix} \ddot{\ell}_\beta \\ \ddot{\ell}_{\sigma^2} \end{bmatrix} = \begin{bmatrix} (\varphi_1 - \mathrm{E}[\varphi_1 \mid \ddot{S}])\mathrm{E}[c \mid \ddot{S}] + (\varphi_2 - \mathrm{E}[\varphi_2 \mid \ddot{S}]) \\ \varphi_3 - \mathrm{E}[\varphi_3 \mid \ddot{S}] \end{bmatrix},$$

where

$$\varphi_1 = -\frac{1}{\sigma^2}\left(\sum_{t=1}^{T-1} y_t - \beta y_0^2\right),$$

$$\varphi_2 = \frac{1}{\sigma^2}\sum_{t=1}^{T} y_{t-1}(y_t - \beta y_{t-1}),$$

$$\varphi_3 = \frac{1}{2(\sigma^2)^2}\left(\sum_{t=1}^{T} (y_t - \beta y_{t-1})^2 + (1-\beta^2) y_0^2\right).$$

Observe that

$$\begin{bmatrix} \ddot{\ell}_{\beta} - \ell^{*}_{\beta} \\ \ddot{\ell}_{\sigma^2} - \ell^{*}_{\sigma^2} \end{bmatrix}$$

$$= \begin{bmatrix} (\mathrm{E}[\varphi_1 \mid y_0, S^*] - \mathrm{E}[\varphi_1 \mid \ddot{S}])\mathrm{E}[c \mid \ddot{S}] + \mathrm{E}[\varphi_2 \mid y_0, S^*] - \mathrm{E}[\varphi_2 \mid \ddot{S}] \\ \mathrm{E}[\varphi_3 \mid y_0, S^*] - \mathrm{E}[\varphi_3 \mid \ddot{S}] \end{bmatrix}$$

is orthogonal to $(\ell^{*}_{\beta}, \ell^{*}_{\sigma^2})'$, from which we can easily verify that the information contained in $(\ddot{\ell}_{\beta}, \ddot{\ell}_{\sigma^2})$ is bigger than the one contained in $(\ell^{*}_{\beta}, \ell^{*}_{\sigma^2})$.

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